

In Class - Mar 23

RJ Cass

$$\begin{aligned}\bar{\mu} &= \frac{1}{3}[E(Y_{Yellow}) + E(Y_{Black}) + E(Y_{Red})] \\ &= \frac{1}{3}[\beta_0 + (\beta_{Black} + \beta_0) + (\beta_{Red} + \beta_0)] \\ &= \beta_0 + \frac{1}{3}\beta_{Black} + \frac{1}{3}\beta_{Red} \\ &= \begin{bmatrix} 1 & 1/3 & 1/3 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_{Black} \\ \beta_{Red} \end{bmatrix}\end{aligned}$$